

KOTA SANDEEP KUMAR

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CAREER OBJECTIVE

Computer Science undergraduate with experience in full-stack development, AI-powered applications, ML-based prediction systems, and cloud technologies. Passionate about building scalable software solutions and seeking a software engineering role where I can contribute while continuously learning.

EDUCATION

Vellore Institute of Technology, Vellore
B.Tech in Computer Science and Engineering

2023 – 2027
CGPA: 7.71 / 10

Sri Chaitanya Junior College
Intermediate (CBSE)

2021 – 2023
Percentage: 70.4%

Sri Chaitanya Techno School
10th (SSC)

2020 – 2021
GPA: 10

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, HTML, CSS

Frameworks & Libraries: React.js, Node.js, Express.js, FastAPI, XGBoost, scikit-learn, Framer Motion, Recharts

Databases: MySQL, MongoDB, Firebase Firestore

Cloud & Tools: Oracle Cloud Infrastructure (OCI), Git, GitHub, Vercel, Render, Uvicorn, VS Code, Linux, Cisco Packet Tracer

Relevant Coursework: Data Structures & Algorithms, Database Management Systems (DBMS), Operating Systems, Object-Oriented Programming (OOP), Software Engineering

PROJECTS

FusionFlowAI – Urban Traffic Congestion Predictor



Next.js / FastAPI / Python / XGBoost / TailwindCSS / Recharts

- Built a smart city AI platform predicting real-time traffic volume at Silk Board Junction, Bengaluru using XGBoost Regressor fused with live weather telemetry (temperature, humidity, and weather condition).
- Implemented a dual ML architecture – Historic Baseline Model and a Weather-Fused Prediction Model – achieving 10–20% accuracy improvement over static historical baselines, evaluated via RMSE, R-squared, MAE, and MAPE metrics.
- Developed an interactive Next.js dashboard with 24-hour comparative charts, hour-by-hour congestion heatmaps, and a model validation table.
- Built a secure Admin Panel with live weather simulation, on-demand model retraining, and database ingestion, all controllable from the browser.
- Deployed FastAPI backend on Render and Next.js frontend on Vercel for continuous 24/7 cloud hosting.

Study AI Desktop Assistant & Command Center



Electron / Node.js / Express.js / Firebase Firestore / Vercel / JavaScript

- Developed a cross-platform desktop application using Electron and Node.js for silent, low-profile screen and audio capture.
- Built and deployed a stateless serverless Express backend on Vercel, integrating Firebase Firestore to manage license keys, HWID tracking, and connection logs.
- Designed a high-performance image preprocessing pipeline with client-side resizing (to 1280px) and JPEG compression, reducing request payload sizes by 95%+ (from 8 MB+ to under 70 KB) to bypass Vercel's 4.5 MB payload limit.
- Implemented stateless API key load balancing across a dynamic pool of Gemini API keys to prevent rate-limiting and quota blocks.
- Created global keyboard hooks for stealth overlays and automated code injection using VBScript and PowerShell scripts.

Hospital Management System



React / TypeScript / Tailwind CSS / Python / FastAPI / MongoDB

- Developed a scalable full-stack web application with a React frontend and a high-performance Python FastAPI backend.

- Designed and integrated an asynchronous MongoDB database using Motor for efficient, non-blocking CRUD operations.
- Implemented secure user authentication and session management using JWT (JSON Web Tokens) across RESTful APIs.

Unisphere



React.js | JavaScript | HTML | CSS | Node.js

- Created a collaborative platform enabling students to share knowledge, clarify doubts, and manage events online.
- Built a responsive React.js frontend and integrated backend functionality using Node.js.
- Implemented discussion forums, event management, and real-time interaction features.

Cryptocurrency Transaction Management System



JavaScript | C# | ASP | Database Systems

- Developed a web-based system to simulate and manage cryptocurrency transactions securely.
- Implemented backend logic using C# and ASP for transaction processing, validation, and data handling.
- Designed interactive frontend components in JavaScript for a seamless user experience.
- Integrated database functionality for storing transaction records and ensuring data consistency.
- Applied basic cryptographic concepts to enhance data integrity and transaction security.

RESEARCH & PUBLICATIONS

“Real-Time Data Fusion for Urban Traffic Prediction”

Co-Author | Submitted to IEEE INDICON 2026 (India Council International Conference) | Paper ID: 2468 | August 2026

Vellore Institute of Technology, Vellore

- Proposed a two-stage ML pipeline fusing real-time weather and traffic data with historical baselines using XGBoost Regressor for accurate urban traffic volume forecasting.
- Demonstrated 10–20% accuracy improvement over static historical baselines, evaluated using RMSE, R-squared, MAE, and MAPE metrics.
- Co-authored with faculty researchers at VIT SCOPE: Sivarangani A. and Priya Ponnuswamy P.

CERTIFICATIONS

- Oracle Cloud Infrastructure – Generative AI Professional
- Google AI Professional Certificate
- be10X – AI Tools Workshop
- Claude Code 101 – Anthropic

